



Slide 1

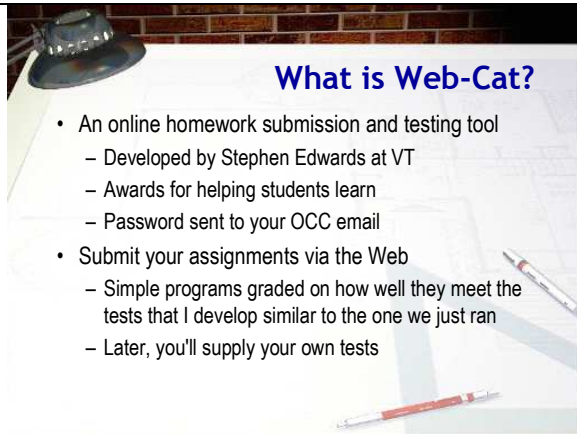
Java Programming 1 Lecture 2F Introducing Web-Cat

Duration: 00:00:29
Advance mode: Auto

Notes:

This is the CS 170 Java Programming 1, Lecture 2F, Introducing Web-Cat.

In the previous lecture, you entered, compiled, ran and tested the Java application, TicTacToeBoardPrinter.java. In this short introduction to Web-Cat, we'll pretend that this was a homework assignment, and you'll see how you would turn it in and how it would be graded.



Slide 2

What is Web-Cat?

Duration: 00:01:09
Advance mode: Auto

- An online homework submission and testing tool
 - Developed by Stephen Edwards at VT
 - Awards for helping students learn
 - Password sent to your OCC email
- Submit your assignments via the Web
 - Simple programs graded on how well they meet the tests that I develop similar to the one we just ran
 - Later, you'll supply your own tests

Notes:

Web-Cat is the program that will be used to grade many of your programming assignments. Web-Cat is an online homework submission and testing tool that was developed by Dr. Stephen Edwards at Virginia Tech. Web-Cat has won several prestigious awards because it's a unique tool that teaches students how to test their own code. For the last several semesters VT has graciously allowed us to use their servers for our CS 170 classes.

The password for your Web-Cat account has already been sent to you via your OCC email. Make sure you save your password, since I don't have any ability to reset it. (If you forget it, though, the Web-Cat site will resend it to your email. And, if you don't like your password, you can change it.)

With Web-Cat, you'll submit your programming assignments via the Web. When we first get started, your assignments will be graded on how well they meet the programming specification. To do this, Web-Cat runs a series of tests similar to the one you just ran (but more sophisticated). Later on, you'll develop your own tests as well.

Simple Testing

- Testing is based on :
 - checking the syntax against the specification
 - running your code with different inputs
 - comparing your output with expected (correct) output
- Most homework assignments are graded
 - 4.5 points given for correct output
 - .5 point given for style using Checkstyle and PMD
 - Comments, indenting, following style guide

Slide 3
Simple Testing
 Duration: 00:01:02
 Advance mode: Auto

Notes:

Web-Cat works by running a set of tests that I've written. These tests:

- Check your syntax against the problem specification. Did you name your class correctly? Do you have the correct methods? Do you have the correct number and type of fields? Basically, I check that the structure of your program is syntactically correct and meets the requirements.
- Then, I check the behavior of your program by running your code with different input (if required) and then comparing your program's output with the output I'd expect if your program was performing correctly.

Most of your homework assignments will be graded on a 5-point scale. Of those 5 points, 90% is given for getting the structure and the behavior correct, and 10% is given for following the style guide. The style portion of your program is automatically checked by a pair of programs named CheckStyle and PMD. These are run automatically by Web-Cat.

Web-Cat via the Web

- Visit: <http://web-cat.cs.vt.edu/>

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Web-Cat via the Web
 Duration: 00:00:44
 Advance mode: Auto

Notes:

Today, we'll look at submitting your assignments via the Web. Next week, after you learn to use the BlueJ IDE, I'll show you how to submit them automatically from within your development environment. To let you gain experience, we'll submit the TicTacToePrinter program you just completed. Your Web-Cat grade for this exercise won't be recorded, though. It's just for practice.

To get started, visit <http://web-cat.cs.vt.edu/>, which brings up the page shown here. To log into the server, click the button that the arrow points to.

Logging In

Notes:

When you reach the login page, enter your user name and your password. The user name will be the same name as your OCC email address, without the @student.occc part. Your password was sent to your OCC email address.

Next, use the drop-down list to change the Institution to Orange Coast College, and click the Login button.

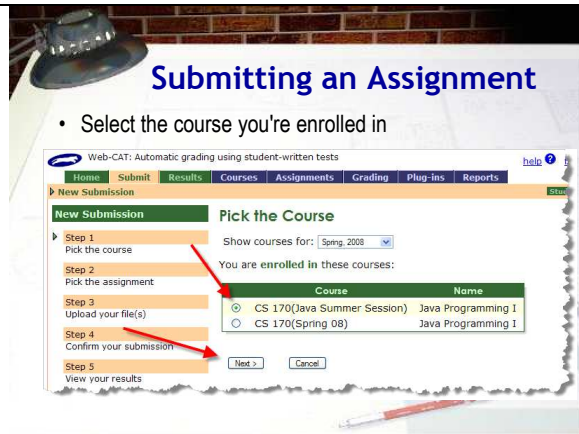
Note that if you've forgotten your password, you can enter your user name and click the Forgot your password hyperlink, and Web-Cat will mail your password to your email address.

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Logging In

Duration: 00:00:37

Advance mode: Auto



Notes:

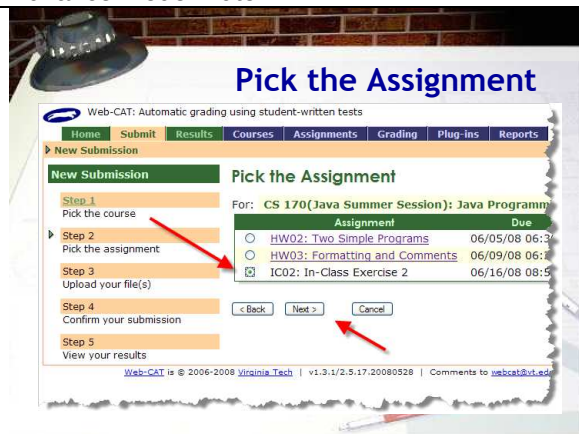
To submit your assignment, choose the Submit tab at the top of the screen, and then select the section you are enrolled in. In the picture, I've shown the classes for Spring and Summer 2008. If you don't see your class, use the "Show courses for" drop-down list to make sure you're looking in the right semester.

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Submitting an Assignment

Duration: 00:00:23

Advance mode: Auto



Notes:

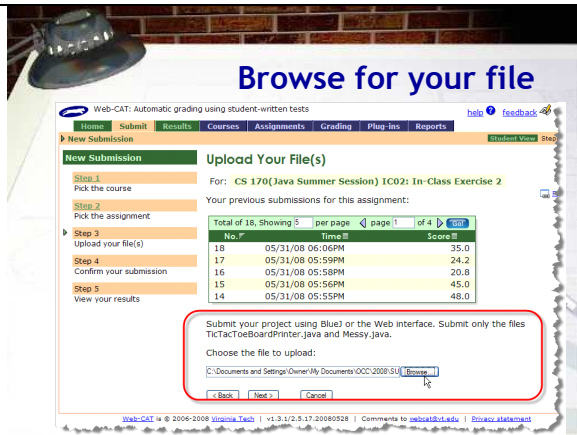
Once you've selected your course and clicked the Next button, you'll see a list of available assignments. For this exercise, choose the IC02 In-class Exercise 2, and click the next button. Note that homework assignments will have a due date, and you won't be able to submit them to Web-Cat after the due date.

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Pick the Assignment

Duration: 00:00:24

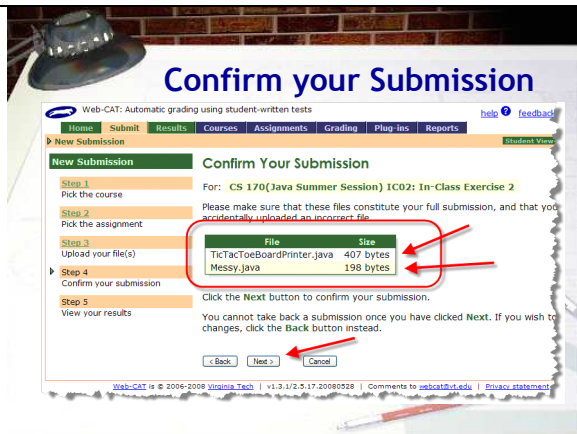
Advance mode: Auto



Slide 8
Browse for your file
 Duration: 00:00:23
 Advance mode: Auto

Notes:

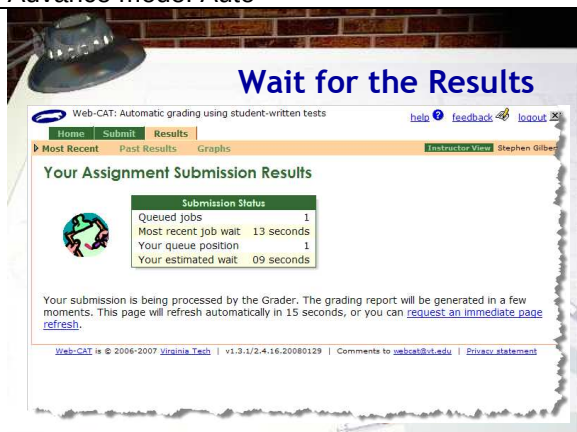
On the next screen you'll find instructions for uploading your file. For this exercise, we're going to upload only the TicTacToeBoardPrinter program (not Messy.java) as in the screen shot. Click the Browse button and when the upload dialog appears, locate the file on your local machine, and click OK.



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Confirm your Submission
 Duration: 00:00:19
 Advance mode: Auto

Notes:

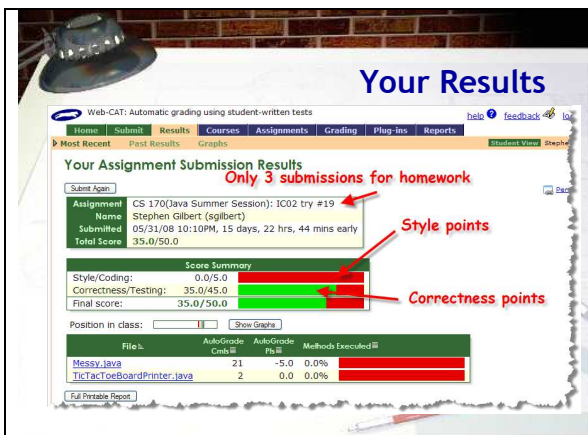
When the file is uploaded, you'll be given a chance to confirm your submission or "back out". Make sure you've uploaded the correct file. In today's exercise, the only file will be TicTacToeBoardPrinter.java. If that's what you see, then click the Next button.



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Wait for the Results
 Duration: 00:00:18
 Advance mode: Auto

Notes:

When you submit your file, it's put in a queue until Web-Cat has an opportunity to run its tests. This usually takes less than a minute. The Web page will automatically refresh once your job has been processed.



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Your Results

Duration: 00:00:44

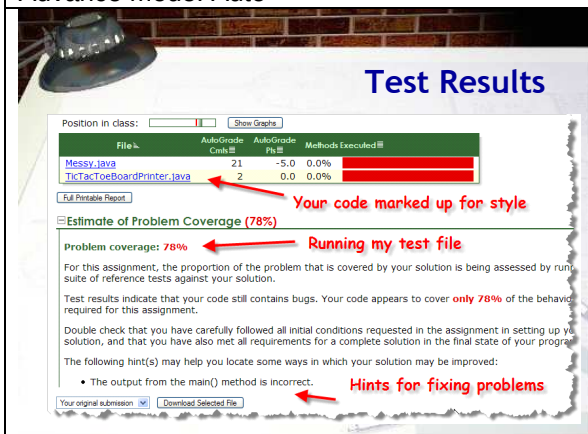
Advance mode: Auto

Notes:

Finally, the results.

First, for this exercise (IC02), you can submit as many times as you want. For your homework, though, you'll only be allowed three submissions. I want you to encourage you to test your program on your local machine before you submit it to Web-Cat.

In the example shown here, you can see that I got 35 points out of 50. For most homework assignments, Web-Cat will be set to report 50 points, but those points will be scaled down to 5 points when recorded in the gradebook. In the score summary section, you can see that I got 0 out of 5 points for style, and 35 out of 45 for correctness.



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Test Results

Duration: 00:00:45

Advance mode: Auto

Notes:

To see what style problems exist, you can click the hyperlink for the file you uploaded, and you'll be shown a listing marked up with the style errors flagged by Web-Cat.

At the bottom of the screen, you'll see the results from running my set of tests. The program I submitted covered 78% of the behavior required by the program specification. If your program doesn't run 100% correctly, I'll also supply hints for fixing the problems as well. In this case, it looks like everything was OK except the output from my main() method was incorrect. Again, this is very similar to the test program that you ran on your local machine.